INCH-POUND
MIL-I-24092/1
4 May 1989

MILITARY SPECIFICATION SHEET

INSULATING VARNISH, SOLVENT CONTAINING, AIR-DRY FOR SPOT PATCHING AND EMERGENCY REPAIRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24092.

REQUIREMENTS:

1. Grade - CA: Clear, solvent containing, air-dry varnish.

2. Class - 130 and 155. The cured varnish shall meet the requirements of the specified thermal class when determined in accordance with MIL-I-24092.

3. Composition - I or II. Composition I, which consists of aromatic solvent, is normally suitable for most varnish treating facilities. However, composition II shall be specified for those varnish treating facilities which are located in areas requiring controlled emission of aromatic hydrocarbons.

4. The varnish shall meet the qualification and quality conformance requirements specified in tables I and II.

AMSC N/A FSC 5970
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TABLE I. Qualification requirements, grade CA.

Characteristics (tests)	Units	Requirements
Storage life		
Sealed container	Months, minimum	12
Unsealed, closed container	Months, minimum	6
Nonvolatiles	Percent	30 to 60
Flash point	°C, minimum	26
Drying time at 110°C	Hours, maximum	3
Viscosity at 20 r/min	Centipoise	200 to 1200
Build, as received	Mil, minimum	1.0
Dielectric strength:		
96/23/50	Volts per mil, minimum	1500
96/23/96	Percent decrease from 96/23/50 value, maximum	35
24/23/water	Percent decrease from 96/23/50 value, maximum	40
Dissipation factor:		
at 23°C	Percent, maximum	0.20
at 150°C	Percent, maximum	2.50
Dielectric constant:		
at 23°C	Maximum	3.5
at 150°C	Maximum	4.5
Hardness	Shore D, minimum	65
Bond strength:		
at 23°C	Pounds force, minimum	5
Thermal endurance:		
Class 130	Temperature index, °C, minimum	130
Class 155	Temperature index, °C, minimum	155

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TABLE I. Qualification requirements, grade CA - Continued.

Characteristics (tests)	Units	Requirements
Hydrolytic stability	Percent decrease in bond strength, maximum	65
Chemical resistance:		
Hydraulic fluid	Percent decrease in bond strength, maximum	50
Lubrication oil		50
Cleaning fluid		50
Distilled water		50
Detergent solution		50

TABLE II. Qualification conformance requirements, grade CA.

Characteristics (tests)	Units	Requirements
Nonvolatiles	Percent of qualification value	± 10
Drying time at 110°C	Hours, maximum	3.5
Viscosity at 20 r/min	Centipoise, percent of qalification value	± 15
Build, as received	Mil, minimum	0.9
Specific gravity	Percent of as-received value	± 3.0
Hardness	Shore D, minimum	60
Bond strength at 23°C	Pounds force, minimum	5

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NOTE:

Part or Identifying Number (PIN). Part number shall be M24092/1-(dash number indicating thermal class). For example, M24092/1-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance.

Custodians:
Army - ER
Navy - SH
Air Force - 20

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
Army - ME
Navy - MC, AS

Preparing activity:
Navy - SH
(Project 5970-0661-01)

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